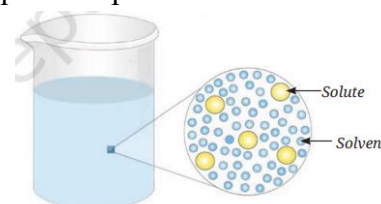


Chapter – 9: The Amazing World of Solutes, Solvents, and Solutions

- Might use – ORS solution – home – some point – life
- ORS – treat dehydration – keep body – hydrated
- Learnt – preparing ORS – home – curiosity – grade 6
- May wonder – homemade ORS – every sip – same taste
- Reason – add sugar, salt – water – form mixture – components – even distribution
- **Predict** – mixture – uniform, non-uniform? – chalk powder – mixed in water – uniform, non-uniform?
- Salt, sugar – mix – water – **uniform mixture**
- Chalk powder, sand, sawdust – mix – water – components – no even distribution
- Such mixtures – **non-uniform mixtures**

What Are Solute, Solvent, and Solution?

- Uniform mixture – salt, sugar + water – solu.
- Solid – mix with liquid – form solution – solid component – **solute** – liquid component – **solvent**
- Solute – dissolves in solvent – form **solution**
 - Solute + Solvent → Solution
- Solution – mixing 2 liquids – not clear – which dissolving which
- Such cases – substance – smaller amt. – solute – larger amt. – solvent
- **Interesting fact** –
 - *Chasni* – sugar syrup – Gulab Jamun – large amt. – sugar (solid) – small amt. – water (liquid)
 - BUT – water – still solvent – sugar – still solute



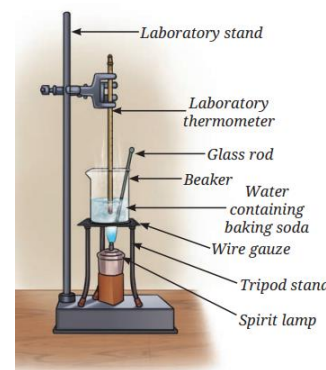
How Much Solute Can a Fixed Amount of Solvent Dissolve?

- Activity –
 - Take – clean glass tumbler – fill half – water
 - Add – 1 spoon salt – stir well – dissolve fully
 - Gradually – add salt – stir – **observe** – how many spoons – until stops dissolving – **record**
- Discussion points
 - How many spoons – before stop dissolving?
 - What indication – capacity – water – dissolve salt?
- Might obs. – salt – complete dissolve – form solution
- Add few more spoons – stage reach – added salt – settle down – reason – water – reached limit
- Solution – more solute – can dissolve – given temp. – **unsaturated solution**
- BUT – solute – stop dissolving – start settling – solution – **saturated solution** – given temp.
- Amt. – solute – fixed qty. – solution (solvent) – **concentration**
- Depend – conc. – **dilute solution** (less solute) – OR – **concentrated solution** (more solute)
- Dil., conc. – rel. terms
- Above act. – solution – 1 spoon salt – dil. – 2 or more spoons – conc.
- **Reflect** – which solution – more conc. – 2 spoons in 100 mL – OR – 4 spoons in 50 mL?
- Max. amt. – solute – dissolve – fixed amt. – solvent – **solubility**

How does temperature affect the solubility of a solute?

- Activity –

- Take – 50 mL – water – glass beaker – **measure** – temp. – lab. thermometer – say – 20 °C
- Add spoonful – baking soda – stir – dissolve complete – continue adding – small amt. – baking soda – stirring – some soda – settle down
- Heat contents – 50 °C – stirring
- Obs. – undissolved soda – dissolves
- Continue adding – small amt. – stirring – some soda – settle down
- Again – heat contents – 70 °C – stirring
- Obs. – undissolved soda – dissolves
- Water – 70 °C – dissolve more soda – water – 50 °C – amt. – dissolved soda – much lesser – 20 °C
- Found – most substances – solubility – increase – temp. increase
- Also say – saturated solution – given temp. – behave as unsaturated – increased temp.
- **Interesting fact** –
 - Water – used as solvent – preparation – medicinal formulations – Ayurveda, *Siddha*, other sys. – medicine
 - Drug formulations – prepared – hydro-alcoholic extracts – herbs
 - Indian sys. – medicine – also refer – use – oils, ghee, milk – solvents – drug formulations – achieve therapeutic benefits



Solubility of Gases

- Many gases – incl. oxygen – dissolve – water
- Oxygen – dissolve – small extent – even then – minute qty. – sustain – aquatic life – plants, fishes, other org.
- Mixture – gases – water – uniform – dissolve evenly
- Obs. – solubility – gases – decrease – temp. – increase
- More oxygen – dissolve – cold water – ensure sufficient oxygen – aquatic life
- Other hand – water – warm up – solubility decrease

Why Do Objects Float or Sink in Water?

- Obs. – some obj. – float – others – sink
- Notice – wash rice – husk particles – float – water surface – rice – sinks – bottom
- Add oil – water – floats
- Gen. belief – obj. – **float** – liquid – lighter – other – **sink** – heavier than liquid
- Wooden stick – iron rod – same size – BUT – iron rod – heavier
- We say – iron – heavier than wood – we refer – special prop. – density – describe – heaviness – obj.

What is Density?

- Imagine – crowded bus – many people – packed together – example – high density – BUT – same bus – only few people – example – low density
- Forest – trees – grow close – each other – dense forest – BUT – trees – far apart – less dense
- **How do scientists define density?**
 - Learnt – matter – possess mass, occupy space

- **Density** – defined – mass in unit vol.
- Density – expressed mathematically –
 - $\text{density} = \frac{\text{mass}}{\text{volume}}$
- Density – independent – shape, size – dependent – temp. pressure
- Pressure – affect density – gases – BUT – effect – solids, liquids – negligible
- Units – density – expressed – depend – units – mass, volume
- Learnt – **SI Units** – **mass, volume** – **kilogram (kg), cubic metre (m³)**
- Hence – **SI Unit** – **density** – **kilogram per cubic metre (kg/m³)**
- Liquids – other units – used – convenience – gram per millilitre (g/mL) – gram per cubic centimetre (g / cm³)
- **Conversion factor for density**
 - $1 \text{ kg/m}^3 = 1000 \text{ g/m}^3 = 1000 \text{ g} / 1000 \text{ L} = 1 \text{ g/L} = 1 \text{ g} / 1000 \text{ mL} = 1 \text{ g} / 1000 \text{ cm}^3$
 - Mass – 1 mL water $\approx$ 1 g – room temp.
 - Measure mass – water – consider – vol. – mL – mass – g
 - 10 mL water $\approx$ 10 g water
 - Suppose – mass – aluminium – 27 g – volume – 10 cm³ – density – 2.7 g/cm³
 - Can say – aluminium – 2.7 x denser – water
 - Express fact – say – rel. density – aluminium w.r.t. water – 2.7 – no. w/o units
 - $\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at same temp.}}$

Determination of density

- Density – determine – measure – mass, volume
- **How to measure mass?**
 - Learnt term – mass – curiosity – grade 6
 - **Mass** – qty. – matter – any obj. – instrument – measure mass – **balance**
 - Must see – various types – balances – shopkeepers use
 - Here – **digital weighing balance** – measure mass
 - Learnt – chapter – exploring forces – Earth – weight, mass – close rel.
 - Activity –
 - Switch ON – digital weighing balance
 - Obs. – initial reading – display
 - Should show – zero – if not – must bring – zero – press tare, reset button
 - Place – dry, clean – watch glass, butter paper – pan – note reading
 - Reset balance – zero – tare, reset button
 - Carefully – place solid obj. – stone – watch glass
 - Note reading – balance – mass – stone – say – 16.400 g
 - Note – mass – liquid – measure – replace watch glass – beaker – pour liquid
 - **A Step Further**
 - Mentioned – chapter 5 – mass, wt. – used interchangeably – everyday lang.
 - BUT – diff. meanings – science – confusion
 - Mass – qty. – matter – units – gram (g) – kilogram (kg)
 - Other hand – wt. – force – Earth attracts – obj. – towards itself – units – Newton (N)
 - Most balances – measure wt. – units – marked – mass scales – show values – grams, kilograms
- **How to measure volume?**
 - Tetra pack – 200 mL buttermilk (*chach*) – meaning?
 - Learnt – curiosity – grade 6 – **volume** – space occupied

- Also know – SI Unit – cubic metres – m^3 – vol. – cube – each side – 1 m – length
- Vol. – smaller obj. – decimetre cube (dm^3) – OR – centimetre cube (cm^3) – cc
- Volume – liquids – litres (L) = 1 dm^3
- Common submultiple – litre – millilitre (mL) = 1 cm^3
- Common apparatus – measure vol. – liquids – **measuring cylinder**
- Narrow transparent cylindrical cylinder – 1 side – open – other side – closed
- Markings – transparent body – indicate vol. – liquid – use it – measure – desired amt. – liquid
- Measuring cylinders – diff. sizes – 5 mL, 10 mL, 25 mL, 50 mL, 100 mL, 250 mL
- Activity –
 - Curiosity – grade 6 – chapter – temp. – learnt – use thermometer – find – smallest reading – do same – measuring cylinder
 - Take cylinder – obs. – note down – max. vol. – measurable – say 100 mL
 - Smallest vol. – measurable – vol. diff. – 2 bigger marks – example – 10 mL, 20 mL
 - Count – smaller divisions – 2 bigger marks
 - Calculate – vol. – 1 small division
- Measuring cylinder – vol. diff. – 10 mL, 20 mL – 10 mL
- No. – divisions – these marks – $10 / 10 = 1 \text{ mL}$
- Smallest value – measuring cylinder – measure – 1 mL
- Smallest vol. – measure – cylinder – depend – capacity
 - 0.1 mL – 10 mL, 25 mL capacity
 - 1 mL – 100 mL capacity
 - 2 mL – 250 mL capacity
 - 5 mL – 500 mL capacity
- Suppose – measure – 70 mL – cannot use – 50 mL – not sufficient – multiple steps – not convenient
- 250 mL, 500 mL – cannot use – smallest vol. – greater than 100 mL – not accurate – measure < 100 mL
- 100 mL cylinder – best choice
- Activity –
 - Place – clean, dry measuring cylinder – flat surface
 - Pour water – measuring cylinder – req. mark
 - If req. – adjust – water level – measuring cylinder – add, remove water – small amt. – dropper
 - Careful obs. – notice water – measuring cylinder – curved surface – **meniscus**
 - Read mark – cylinder – coincide – bottom – meniscus – water, colourless liquids
 - Make sure – eyes – level – bottom – meniscus
 - Once reach – req. amt. – 50 mL – transfer – another container
- Coloured liquids – cylinder mark – coincide – top – meniscus



• Determining volume of solids with regular shapes

- Activity –
 - **Collect** – various obj. – cuboid shapes – notebook, shoebox, dice
 - Measure – length (l), width (w), height (h) – obj. – scale
 - Suppose – notebook – $l = 25 \text{ cm}$ – $w = 18 \text{ cm}$ – $h = 2 \text{ cm}$
 - **Calculate** – vol. – following formula
 - Volume = $l \times w \times h$
 - Volume = $25 \times 18 \times 2 = 900 \text{ cm}^3$ – record

- **Determining volume of objects with irregular shapes**

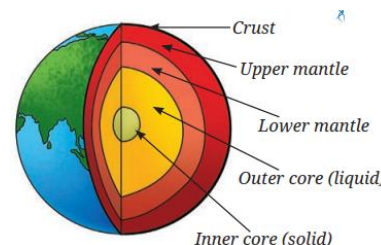
- Imagine – obj. – stone – no regular shape
- Calculate density – main challenge – find vol.
- How to find vol.? – irregular shape
- Activity –
 - Collect – various obj. – surroundings – stone, metal keys, etc.
 - Fill – measuring cylinder – water – any vol. – say 50 mL – record – initial vol.
 - Tie obj. – stone – thread – lower it – measuring cylinder
 - Record – final vol. – rise in level – 55 mL
 - Subtract – initial from final vol. – vol. – obj. – record
- Note – values – vol. – units – mL – written – eq. values – units – cm³ – solids

- **Let us calculate density**

- Density – calculate – use formula –
 - $\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{16.400 \text{ g}}{5 \text{ cm}^3} = 3.28 \text{ g/cm}^3$

- **Let us dig dipper**

- Planet Earth – several layers – crust, upper mantle, outer core, inner core – each – fixed range – density
- Outermost layer – crust – lightest – density increase – move toward centre
- Move deeper – Earth – pressure, temp. – increase – materials – heavier, compact
- **Interesting fact –**
 - Ancient times – before large ships – people – bamboo, wooden logs – travel – rivers, seas
 - Bamboo – light, hollow – float easy
 - People – tie bamboo – make rafts, small boats – fishing, trading, crossing water
 - Wooden logs – strong trees – hollowed out – make boats, rafts
 - Simple boats – local materials – imp. – move around – connect places
 - Even to date – similar traditional boats – bamboo, wood – used some regions – transport – BUT also – tourist attraction



Effect of temperature on density

- Density – substance – decrease – heating – increase – cooling
- Explained – basis – learnt – chapter – particulate nature of matter
- Temp. increase – particles – move away, spread – result – vol. increase – BUT – no change – mass
- Density = mass / vol. – heating – vol. increase – density decrease
- This reason – hot air – less dense – move up – hot air balloon – same principle

Effect of pressure on density

- Pressure – diff. effect – diff. state of matter
- Gases – increasing pressure – particles – move close – result – vol. decrease – density increase
- Liquids – nearly incompressible – pressure – small effect
- Solids – particles – very close – even less effect – change in density – negligible
- **Interesting fact –**
 - Ice – floats on water – lighter than liquid water
 - Water – special prop. – density – highest – 4 °C

- Temp. drops – water turns to ice – 0 °C – structure change – particles rearrange – take more space
- Same amt. – water – larger vol. – density decrease – result – ice – lighter – floats
- Imp. – animals – lakes, oceans – floating ice – warm water beneath – fish, other creatures – survive – extreme cold conditions